WfExS-backend in the WRROC world?

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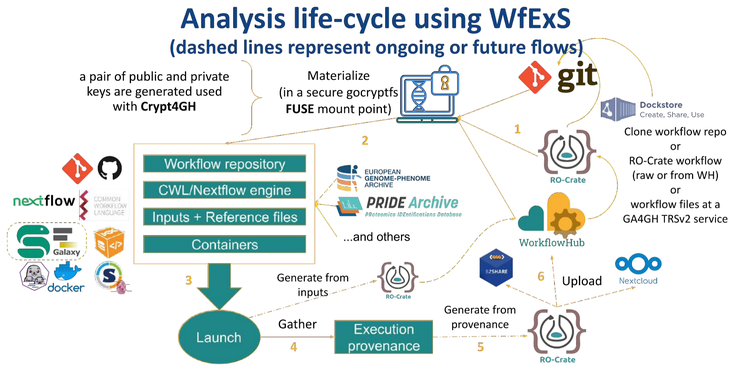
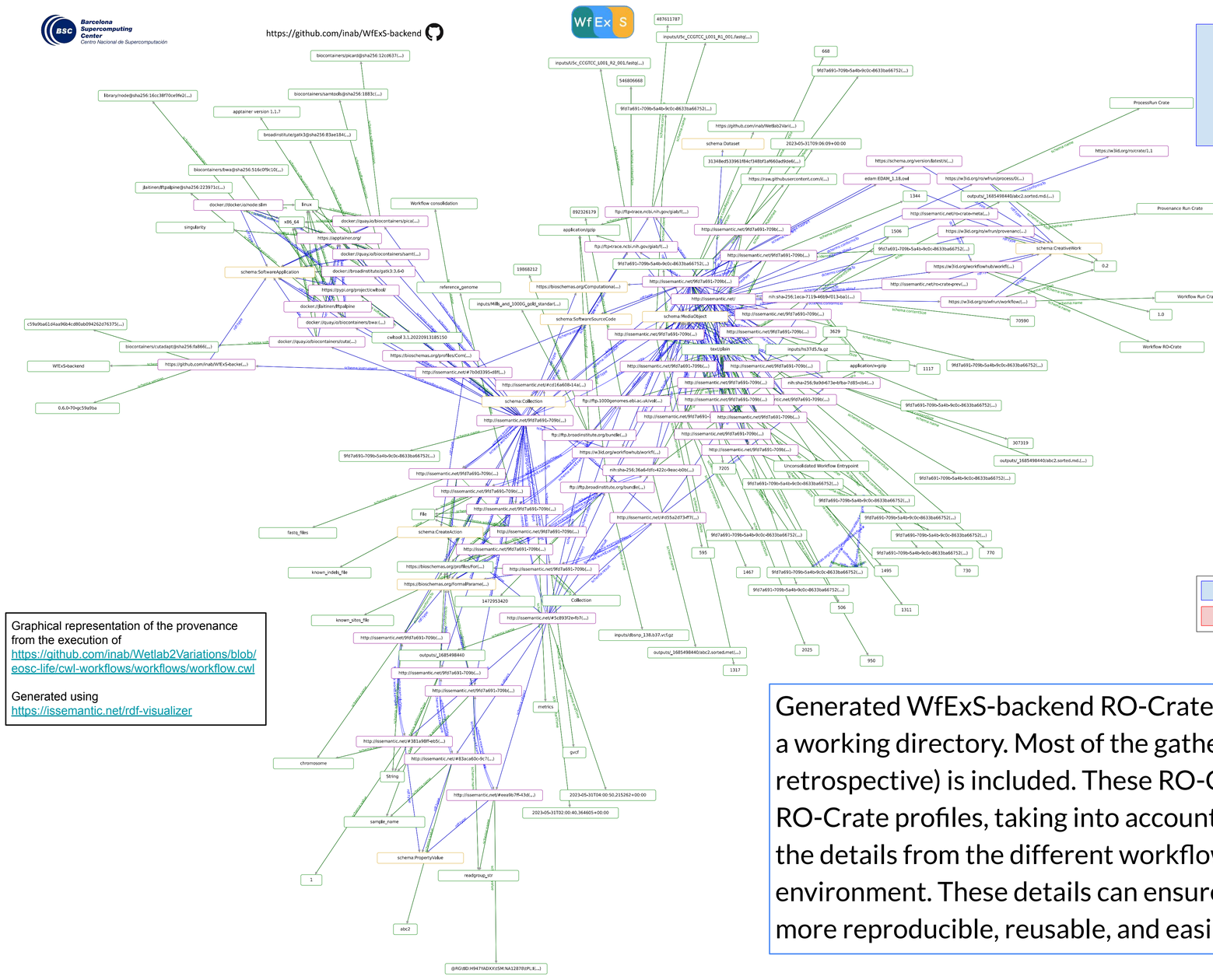
Workflow Execution Service Backend (WfExS-backend) high-level orchestrator has as one of its targets to run scientific workflows reproducibly. A workflow execution lifecycle starts through the acquisition of the workflow from its repository, its analysis in order to learn about the workflow language and dependencies. Such dependencies are the minimal version of the workflow engine needed to successfully run the workflow, additional subworkflows and the software containers needed by the workflow and its dependencies.



The staging of a local workflow execution also needs to obtain the inputs, identified by URLs and PIDs, from a distributed data scenario. Although the inputs can be cached, when a local execution is required to be secure, the working directory is encrypted using a FUSE filesystem like GoCryptFS, and the inputs are not cached.

WfExS-backend can consume and produce RO-Crates describing scientific workflows (where the workflow and its resources are available) and workflow executions (description of inputs, containers, etc...).



Graphical representation of the provenance from the execution of https://github.com/inab/Wetlab2Variations/blob/eosc-life/cwl-workflows/workflows/workflow.cwl

Generated using https://issemantic.net/rdf-visualizer

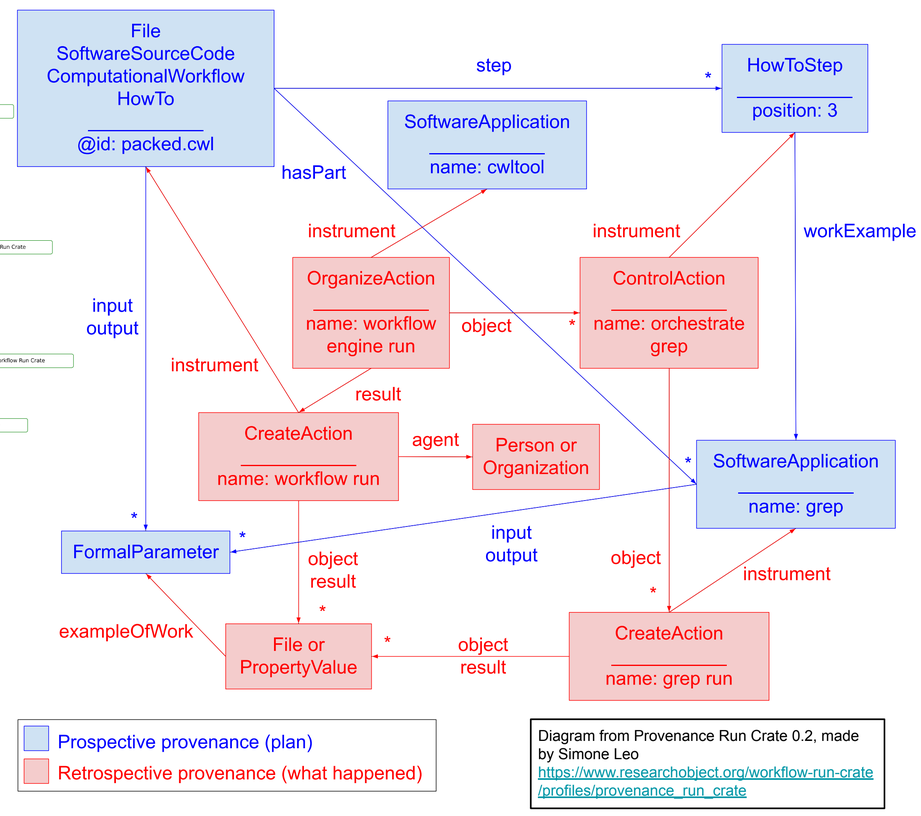


Diagram from Provenance Run Crate 0.2, made by Simone Leo https://www.researchobject.org/workflow-run-crate/profiles/provenance_run_crate

Generated WfExS-backend RO-Crates contain the details from previously staged executions in a working directory. Most of the gathered, associated metadata (both prospective and retrospective) is included. These RO-Crates should follow the different Workflow Run RO-Crate profiles, taking into account the limitations of the WfExS-backend by gathering all the details from the different workflow engines, their instantiations and the computing environment. These details can ensure that properly documented workflow instantiations are more reproducible, reusable, and easily understood by the scientific community.

Included metadata

- Original workflow (and dependencies) location and fingerprints.
- Materialized workflow.
- Containers potentially used, plus their signature and fingerprints.
- Workflow language, and used engine version.
- Original input datasets identity or location plus their fingerprint.
- Generated outputs fingerprints.

Optionally includable

- A copy of the gathered inputs.
- A copy of the gathered software containers.
- A copy of the generated outputs.
- Public identifier of the exported output results (WIP).
- Graphical representation of the workflow (WIP).

Gathered but not included (yet)

- Executed internal workflow steps.
- Relationship between executed workflow steps.
- Intermediate results.
- Execution metrics gathered by the workflow engine, like CPU and memory usage of each of the steps (GPU metrics are not gathered).

WfExS-backend supports both cwltool and Nextflow workflow engines with current efforts to incorporate snakemake and Galaxy. WfExS-backend was initially developed in EOSC-Life as a demonstrator on sensitive data analysis, it is being developed and used in iPC, EJP-RD, BY-COVID and EuroScienceGateway projects, and is now part of the Service Delivery plan of INB/ELIXIR-ES.

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